Chapter 4

Biostatistics and Data Analysis

Contents

- **Measures of dispersion**
  - Range
  - Quartiles
  - IQR
  - Variance

Measures of Dispersion

- A **measure of dispersion** shows how much the data values differ or vary from each other — that is, how **spread out** the data are.

Key Points:

- If all the values are the same → There is no dispersion.
- If all the values are different → There is dispersion.
- If the values are close to each other → Dispersion is small.
- If the values are widely scattered→ Dispersion is large.

Measures of central tendency

Measures of Variability

- **Main Measures:**

1. **Range**

   → The difference between the **largest** and **smallest** values. ***(Range = Maximum – Minimum)***

2. **Quartiles**

   → Divide a set of data into **four equal parts (quarters)**.

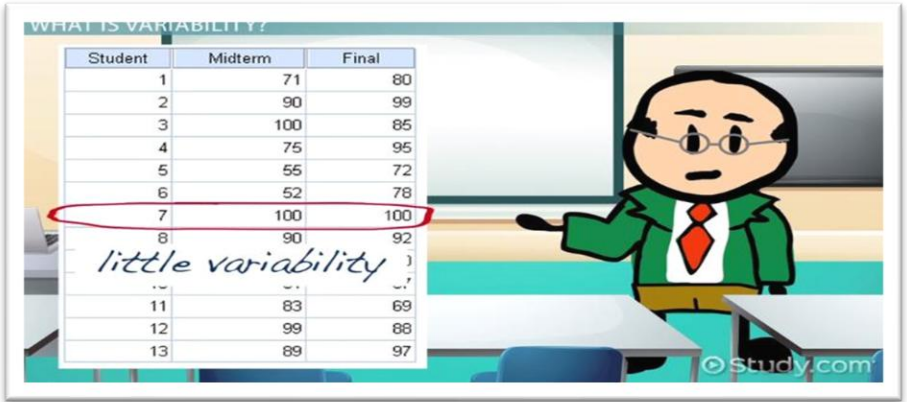
3. **Interquartile Range (IQR)**

   → The difference between the **third quartile (Q3)** and the **first quartile (Q1)**. ***(IQR = Q3 – Q1)***

4. **Variance**

   → Shows **how far the numbers are spread out** from the mean.

(Larger variance = greater spread of data)



Range

- The Range is the difference between the largest value (Max) and the smallest value (Min).
- Range (R) = Max − Min
- Range =Largest value - Smallest value
- Note: Range concern only onto two values
- Find Range?

Example 1: Data: 43, 66, 61, 64, 65, 38, 59, 57, 57, 50.

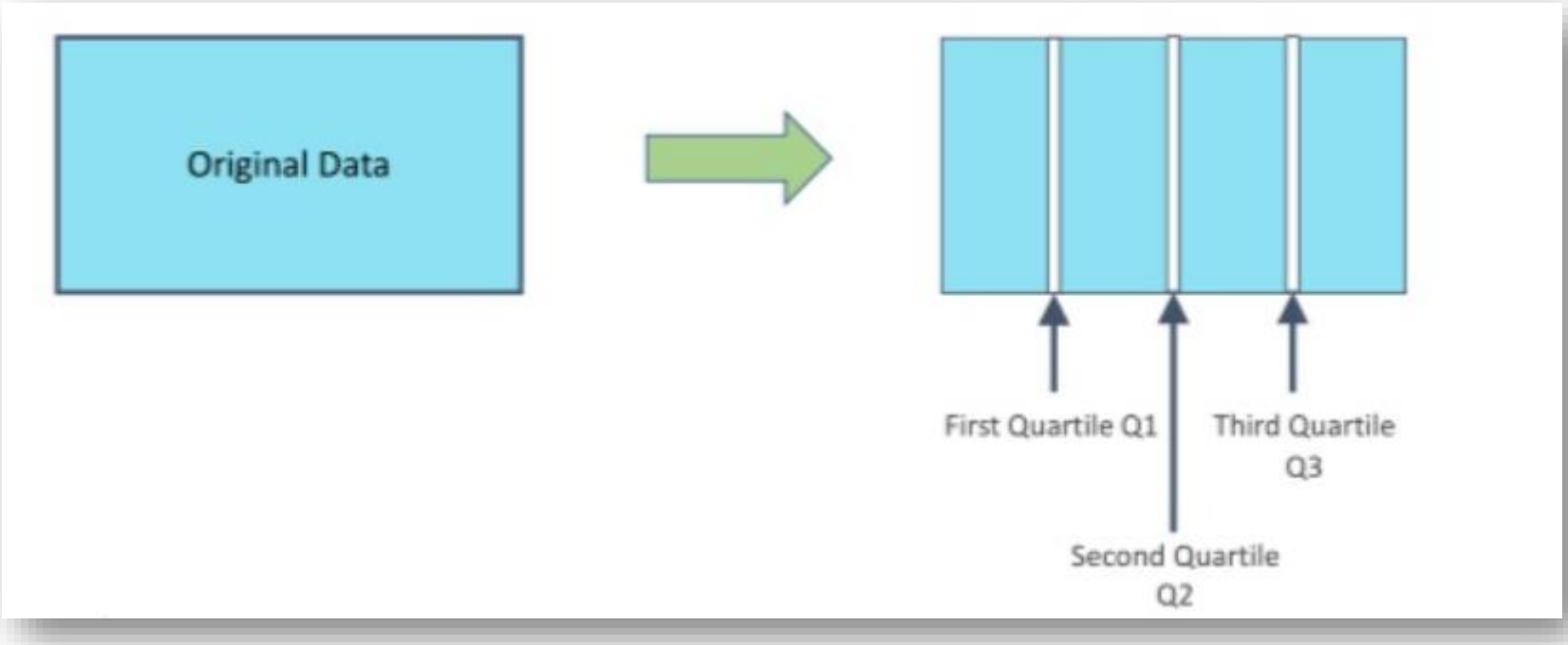
Range = 66 − 38 = **28**

Example 2 Data: 26, 25, 35, 27, 29, 29.

Range = 35 − 25 = **10**

Quartile

- Quartile divides a list of numbers into four quarters.
- The **three quartiles** (Q1, Q2, Q3) divide the data values at the three points that splits the data set into four equal **quarters** (groups).



Quartile

Types of Quartiles

1. First Quartile (Q1 or 25th Percentile)

- Also called the **Lower Quartile**.
- It marks the point below which **25% of the data values** lie (or above which 75% of the data values fall)..
- Represents the boundary between the lowest quarter and the rest of the dataset.

2. Second Quartile (Q2 or 50th Percentile)

- Also called the **Median**.
- It divides the dataset into **two equal halves**.
- Represents the middle value of the ordered data.

3. Third Quartile (Q3 or 75th Percentile)

- Also called the **Upper Quartile**.
- It marks the point below which **75% of the data values** lie (or above which 25% of the data values fall).
- Represents the boundary between the top quarter and the rest of the dataset.

Quartile

How to Calculate Quartiles

1. Arrange data in ascending order.
2. Determine the number of values (n).
3. Calculate quartile positions:

- Q1: (n+1) x 25%
- Q2: (n+1) x 50%
- Q3: (n+1) x 75%

4. Find the corresponding values at these positions.

Quartile

Example No. 1

Calculate **Q1, Q2, Q3** for the given dataset. (3, 4, 8, 5, 10, 9, 1, 4, 6, 12, 2)

- **Step 1: Arrange in Ascending Order**

  1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12

- **Step 2: Number of Values**

  n = 11

- **Step 3: Quartile Positions**

$$Q1 = (n+1) \times 25\% = 12 \times 0.25 = 3 \Rightarrow \textit{3rd value}$$

$$Q2 = (n+1) \times 50\% = 12 \times 0.50 = 6 \Rightarrow \textit{6th value}$$

$$Q3 = (n+1) \times 75\% = 12 \times 0.75 = 9 \Rightarrow \textit{9th value}$$

- **Step 4: Find the Values**

  3rd value = **3** → $Q1$

  6th value = **5** → $Q2$

  9th value = **9** → $Q3$

1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12

Quartile

Example No. 2

Calculate **Q1, Q2, Q3** for the given dataset. (3, 4, 8, 5, 10, 9, 1, 4, 6, 12, 2, 11)

- **Step 1: Arrange in Ascending Order**

  1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 11, 12

- **Step 2: Number of Values**

  n = 12

- **Step 3: Quartile Positions**

- $Q1 = (n + 1) \times 25\% = 13 \times 0.25 = 3.25$
- $Q2 = (n + 1) \times 50\% = 13 \times 0.50 = 6.5$
- $Q3 = (n + 1) \times 75\% = 13 \times 0.75 = 9.75$

1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 11, 12

- **Step 4: Find the Values**

- Q1 → Between 3rd value (3) and 4th value (4): (3 + 4) ÷ 2 = **3.5**
- Q2 → Between 6th value (5) and 7th value (6): (5 + 6) ÷ 2 = **5.5**
- Q3 → Between 9th value (9) and 10th value (10): (9 + 10) ÷ 2 = **9.5**

The interquartile range (IQR)

- The interquartile range (IQR) is the difference between the third and first quartiles.

$$\mathbf{IQR} = Q_3 - Q_1$$

- It is a measure of dispersion.
- A large IQR indicates a large amount of variability among the middle 50 percent of the relevant observations,
- A small IQR indicates a small amount of variability among the relevant observations.

IQR = Q3 – Q1

1, 2, 3, 4, 4, 5, 6, 8, 9, 10, 12

IQR = 9 – 3

= 6

The interquartile range (IQR)

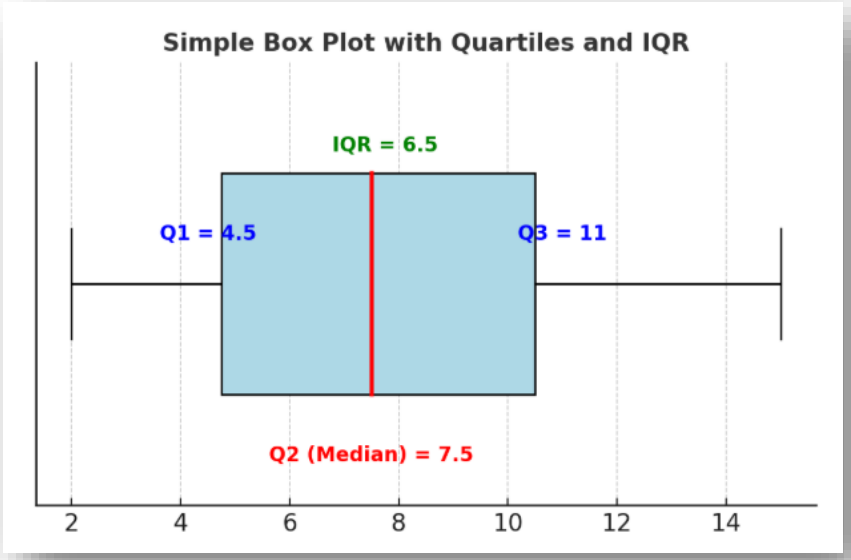
Suppose we have students' test scores, Calculate IQR.

2, 4, 5, 7, 8, 10, 12, 15

- **Q1** = 4.5 (median of lower half → between 4 and 5)
- **Q2 (median)** = 7.5 (middle of all values → between 7 and 8)
- **Q3** = 11 (median of upper half → between 10 and 12)

- IQR = Q3−Q1 = 11−4.5 = 6.5

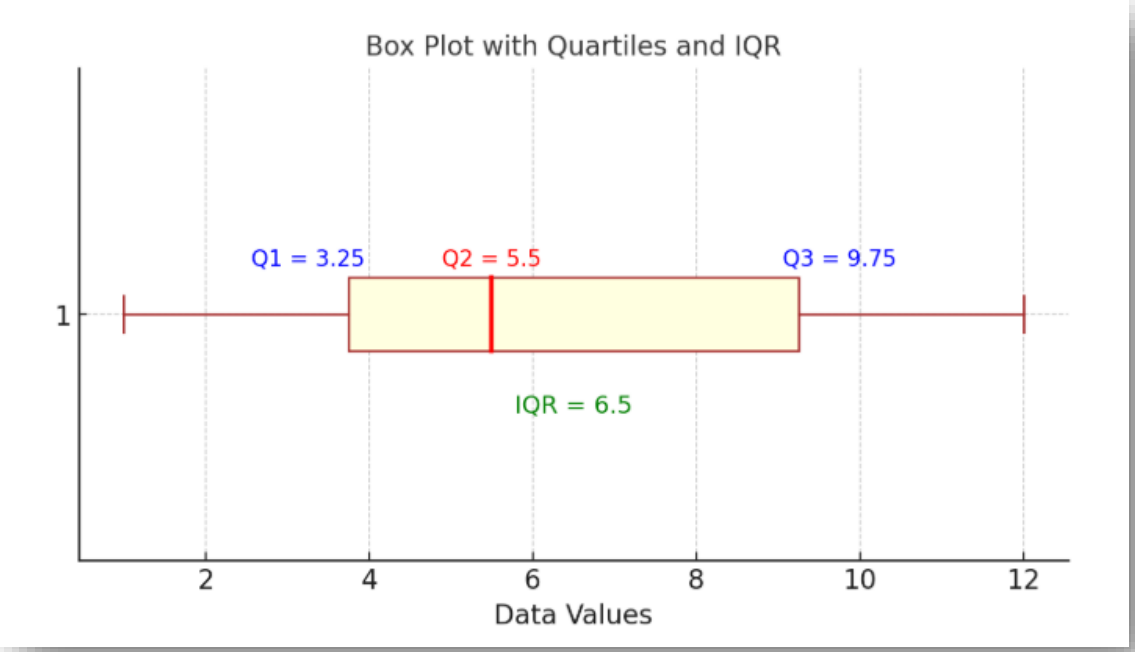
- Here's the simple, **labelled box plot** for students:
  - **Q1 = 4.5** (left edge, blue)
  - **Q2 = 7.5** (median, red)
  - **Q3 = 11** (right edge, blue)
  - **IQR = 6.5** (spread of the box)



The interquartile range (IQR)

Here's another example of **box plot with Q1, Q2, Q3, and IQR labelled**:

- **Q1 = 3.25** (blue, left edge of box)
- **Q2 = 5.5** (red, median line)
- **Q3 = 9.75** (blue, right edge of box)
- **IQR = 6.5** (spread of the box)



Variance

- The variance is one of the most important measures of dispersion.
- The variance is a measure that uses the mean as a point of reference.
- The variance of the data is small when the observations are close to the mean.
- The variance of the data is large when the observations are spread out from the mean.
- The variance of the data is zero (no variation) when all observations have same value.

Variance

- There are two types of variance.
  - Population variance
  - Sample variance

Variance

Sample Variance (s^2)

- When you only have **sample** from population, then its Variance is calculated as:

$$s^2 = \frac{\sum(x_i - \bar{x})^2}{n-1}$$

- s^2 = Sample Variance
- x_i =each data value
- $\bar{x}$ =sample mean
- n= number of sample values
- Divide by **(n − 1)** instead of n→ this is called **Bessel's correction**.

Variance

Population Variance (σ^2)

- When you have data for the **entire population** (everyone, not just a sample):

$$\sigma^2 = \frac{\sum(x_i - \mu)^2}{N}$$

  - σ^2 = Population Variance
  - x_i =each data value
  - μ= population mean
  - N= total number of population values
  - Divide by **N** (since you know the whole population).

Steps for Finding Sample Variance & Sample Standard Deviation

1. Find the mean of the sample data set. $\bar{x} = \frac{\Sigma x}{n}$

2. Find deviation of each entry. $x - \bar{x}$

3. Square each deviation. $(x - \bar{x})^2$

4. Add to get the sum of the deviations squared. $\Sigma(x - \bar{x})^2$

5. Divide by $n - 1$ to get the **sample variance**. $s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$

6. Find the square root to get the **sample standard deviation**. $s = \sqrt{s^2}$

Steps for Finding Population Variance & Population Standard Deviation

1. Find the mean of the Population data set. $\mu = \frac{\sum X}{N}$

2. Find deviation of each entry. $(X - \mu)$

3. Square each deviation. $(X - \mu)^2$

4. Add to get the sum of the deviations squared. $\sum (X - \mu)^2$

5. Divide by N to get the **Population variance**. $\sigma^2 = \frac{\sum (X - \mu)^2}{N}$

6. Find the square root to get the **Population standard deviation**. $\sigma = \sqrt{\sigma^2}$

Sample Variance and Sample Standard Deviation

Sample Variance

$$s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1}$$

Sample Standard Deviation

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma(x - \bar{x})^2}{n - 1}}$$

Population Variance and Population Standard Deviation

Population Variance

$$\sigma^2 = \frac{\Sigma(x-\mu)^2}{N}$$

Population Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\Sigma(x-\mu)^2}{N}}$$

THANK YOU